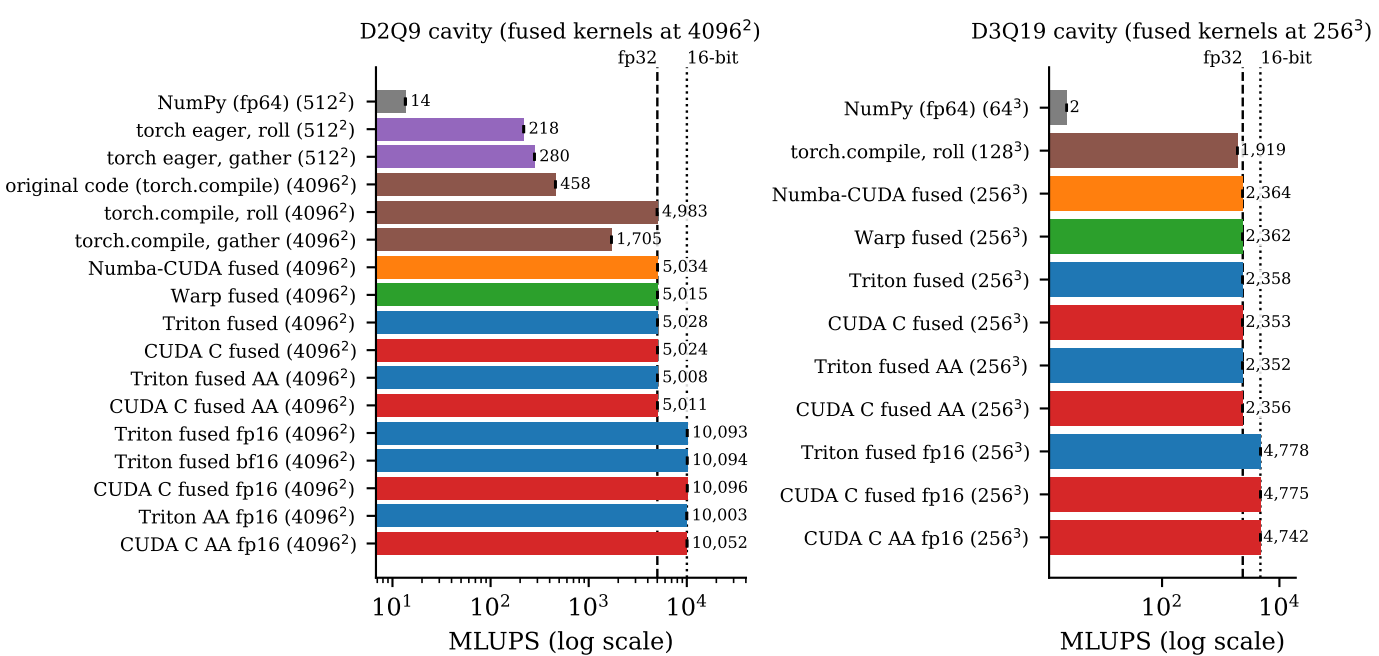
D2Q9 cavity (fused kernels at 4096²)
fp32
16-bit
NumPy (fp64) (512²)
14
torch eager, roll (512²)
218
torch eager, gather (512²)
280
original code (torch.compile) (4096²)
458
torch.compile, roll (4096²)
4,983
torch.compile, gather (4096²)
1,705
Numba-CUDA fused (4096²)
5,034
Warp fused (4096²)
5,015
Triton fused (4096²)
5,028
CUDA C fused (4096²)
5,024
Triton fused AA (4096²)
5,008
CUDA C fused AA (4096²)
5,011
Triton fused fp16 (4096²)
10,093
Triton fused bf16 (4096²)
10,094
CUDA C fused fp16 (4096²)
10,096
Triton AA fp16 (4096²)
10,003
CUDA C AA fp16 (4096²)
10,052
10¹
10²
10³
10⁴
MLUPS (log scale)
D3Q19 cavity (fused kernels at 256³)
fp32
16-bit
NumPy (fp64) (64³)
2
torch.compile, roll (128³)
1,919
Numba-CUDA fused (256³)
2,364
Warp fused (256³)
2,362
Triton fused (256³)
2,358
CUDA C fused (256³)
2,353
Triton fused AA (256³)
2,352
CUDA C fused AA (256³)
2,356
Triton fused fp16 (256³)
4,778
CUDA C fused fp16 (256³)
4,775
CUDA C AA fp16 (256³)
4,742
10²
10⁴
MLUPS (log scale)